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RESEARCH PAPER

Adult acquired major upper limb amputation in Norway: prevalence, demographic features and amputation specific features. A population-based survey

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Abstract

Purpose. To estimate the prevalence of adult acquired major upper limb amputation in Norway. To describe this amputee population regarding demographic features and amputation specific features. To compare our data to data collected internationally.

Method. Population-based cross-sectional study on adult upper limb amputees with acquired limb loss through or proximal to the radio-carpal joint. Patients were found in the databases of the two companies in Norway that make upper limb prostheses and in the medical records of three of the largest Norwegian hospitals. Data were collected by postal questionnaires.

Results. We estimated a population prevalence of 11.6 per 100,000 adults ($n = 416$). Our survey was not 100% comprehensive and the estimate is conservative. The amputees were predominantly men with traumatic, unilateral, distal amputations at a young age. There were significant gender- and amputation level differences in cause. Most amputees had used prostheses. About four in ten were in paid employment.

Conclusions. Our findings are mainly consistent with earlier studies from other countries. Implications of our findings related to the planning of future health care for these patients are outlined, including suggestion of regional multidisciplinary rehabilitation emphasising occupational rehabilitation and focus on preventive measures. Potential areas of follow-up are suggested.

Keywords: *Arm, amputee, amputation, upper limb, Norway*

Introduction

Arms and hands are used for most activities of daily living, and are also important for communication and affection [1,2]. Thus, the loss of an upper limb has an extensive effect on people's lives, resulting in a major restriction of function, sensation and appearance [3]. Major amputation (through or proximal to the wrist) causes a total loss of grip on the affected side and requires comprehensive, multidisciplinary rehabilitation [4]. The importance of optimal rehabilitation of upper limb amputees (ULAs) is emphasised by the fact that upper limb loss typically results from traumatic injury in relatively young, healthy individuals who should expect a normal life span with many years of productive employment and active contribution to society [5,6]. The mean age at amputation varies between 20 and 36 years in studies that look specifically at upper limb loss, the majority stating an age at amputation in the early or mid-twenties [3,5,7–14].

The body of knowledge of acquired ULAs is however small, and existing studies suffer from a host of methodological problems. Especially, there is a paucity of population-based studies. Prevalence estimates are scarce. Methodological problems include different study populations and lack of standardised case-definitions and standardised methods for case ascertainment [15]. Some studies look at

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data from prosthesis fitting centres only, thus excluding the non-wearing proportion of the amputee population [16]. The use of operative reports and medical records may lead to an underreporting of cases, whereas hospital discharge records have the potential for counting multiple amputations in the same person, leading to an overestimation of incidence. Different studies calculate rates as amputations or amputees, making comparisons difficult. These differences in methodology may explain why incidence rates of acquired amputation are found to vary greatly between and within nations [15]. For instance, reported incidence estimates for upper limb loss in the Nordic countries vary from 0.08 per 100,000 per year to 150 per 100,000 per year [17–21]. These are neighbouring countries with similar structure regarding industrialisation, industrial safety legislation and health care, and one would expect their incidence and prevalence of acquired limb loss to be similar.

In earlier literature, prevalence estimates are scarce. According to a report by SINTEF Unimed Rehab in 1996, 417 Norwegian acquired ULAs were registered as users of arm prostheses at the Norwegian prosthetic suppliers Norwegian Technical Orthopaedics AS (NTO AS) and Sophies Minde Orthopaedics AS (SMO AS), including partial hand amputees and children. This number was not corrected for duplicates (amputees registered at both prosthetic suppliers). An estimate of the number of acquired arm amputees not wearing a prosthesis (non-users) was not given [21]. In 2005, Ziegler-Graham et al. estimated a prevalence of acquired major upper limb loss in the USA of 41,000 Americans aged 0–85 years, i.e. about 13.5 per 100,000 [22]. A few population-based or regional studies have been conducted in other countries such as Saudi Arabia [7], Korea [23], Burma [24], Kenya [25] and Malaysia [26] but these countries are in many aspects very different from Norway. As noted by Al-Turaiki and Al-Falahi in 1993, 'The studies in Asia had great similarities to each other, as trauma and disease incidence occur in similar patterns, whereas the patterns are different in America, Denmark, Finland and Australia' [7]. Thus, Norwegian data should primarily be compared to data from other western countries.

In Europe, national arm amputation statistics exist in England [27,28] and Denmark [19,29]. In Norway, there are no reliable national statistics regarding incidence and prevalence of acquired upper limb loss [21]. In Sweden, a pilot quality registry of ULAs (acquired and congenital) was started in 1998, but it was never as comprehensive as intended and eventually terminated [30].

The majority of amputation literature deals with lower extremity amputation [31–34]. Many publications look at both upper and lower limb amputations or include congenital limb loss, often making it difficult to abstract specific data on acquired upper limb loss [7,8,15,16,20,22–27,35–40]. Several studies include finger- and partial hand amputation, these patients then usually outnumber patients with more proximal limb loss [15,17,20,22,36]. Given the unique nature and functions of the hand and the upper limb, the rehabilitational needs of major ULAs are likely to be very different from those of lower limb amputees, and also from those of partial hand amputees and finger amputees who have part of their grip and/or wrist motion intact. Furthermore, ULAs differ from lower limb amputees in several important aspects such as mean age at amputation, average number and pattern of co-morbidities and cause of amputation [5,41,42]. Thus, pooled knowledge of several groups of amputees may not be applicable to major ULAs.

Currently, the rehabilitation of ULAs in Norway is not standardised. Its organisation differs between the five sites offering specialised upper limb rehabilitation and amputees are not routinely referred to rehabilitation or prosthetic fitting at discharge from the surgical ward. Successful planning and organising of future health services for these patients requires specific knowledge of both upper limb amputation prevalence and upper limb amputee population characteristics. This knowledge will be essential in addressing issues like the geographic organisation of upper limb rehabilitation and the amount of resources to be allocated to this patient group. Information on the causes of amputation may also guide public health measures to prevent traumatic limb loss. Given the paucity of and methodological problems related to existing literature, the need for further information is evident.

Purpose of study

The aim of this cross-sectional study was to estimate the population prevalence of adult acquired major upper limb amputation in Norway, and to describe this amputee population with regard to demographic features and amputation specific features. Furthermore, we aimed to compare our data to data collected in other studies internationally to ascertain whether the Norwegian major ULA population is comparable to the major ULA populations in other western countries. Conducting a population-based survey pooling several different sources of inclusion and applying clear case-definitions, we aimed to provide specific and comprehensive data as an important addition to the relatively small and methodologically variable existing body of knowledge of major ULAs.

Method

Design and inclusion and exclusion criteria

We performed a cross-sectional study. Inclusion started on 1 October 2006 and was terminated on 30 May 2008. Inclusion criteria for the estimation of prevalence were: adult (≥18 years of age at 1 October 2006, i.e. born 1 October 1988 or before), acquired upper limb loss, level of amputation through or proximal to the radio-carpal joint (major ULA) and resident in Norway at the time of the survey. For the questionnaire study, an extra inclusion criterion was mastering spoken and written Norwegian and an exclusion criterion was severely reduced cognitive function or physical condition.

The inclusion process

In order to estimate the prevalence of adult acquired major ULAs in Norway, we used the databases of the two companies in Norway that make upper limb prostheses, NTO AS and SMO AS. The initial electronic searches were manually verified and supplemented by manual examination of the databases according to the inclusion criteria. Amputees not wearing a prosthesis (non-users) were found through a search for relevant ICD-10 diagnostic codes and NCSP surgical procedure codes in the electronic medical records of three of the largest Norwegian hospitals: Oslo University Hospital Rikshospitalet, St. Olavs Hospital (Trondheim) and Haukeland University Hospital (Bergen). Haukeland University Hospital and St. Olavs Hospital are regional hospitals in the Western Norway and Central Norway health regions with a total of about 1,530,000 inhabitants. Haukeland University Hospital treats approximately 600,000 patients per year, and is the largest hospital in Norway in terms of number of patients [43–45]. Oslo University Hospital Rikshospitalet is a highly specialised university hospital with national responsibilities within complicated treatments. During the last 20 years, all replantation surgery in Norway has been performed here (earlier also in Bergen and Trondheim). In 2007, Oslo University Hospital Rikshospitalet treated approximately 323,000 patients [46]. Codes in operative reports, inpatient and outpatient hospital discharge records were examined. We also performed a manual search for relevant ICD-10 diagnostic codes in the non-electronic medical records at the Department of Occupational Therapy at Oslo University Hospital Rikshospitalet and the Department of Orthopaedic Rehabilitation at Haukeland University Hospital. In addition, we went through handwritten lists of NCSP surgical procedure codes for patients treated on the surgical ward at St. Olavs Hospital. All ICD-10-code and NCSP-code diagnoses were verified by manual control of medical records or by questionnaire response. Patients with codes that could not be manually verified were excluded, as were several patients who had been miscoded as major ULAs. To minimise the risk of missing data due to miscoding of major ULAs as other diagnoses, we included several additional ICD-10 and NCSP codes related to amputation complications in our search. A cross-check was performed and no additional major ULAs were found based solely on these additional codes. During inclusion, lists of potential participants from the different sources were pooled and duplicates removed sequentially based on individual National Insurance numbers. The final list was checked against the National Population Register (DSP) to exclude patients who had died or emigrated. Finally, some amputees included in the prevalence estimate were excluded from the questionnaire study according to the questionnaire study-specific inclusion and exclusion criteria. Details of the inclusion process are shown in Figure 1.

The questionnaire study

In order to describe the characteristics of the Norwegian adult acquired major upper limb amputee population, we used postal questionnaires returned by pre-paid mail. Submission of the questionnaire was also consent to participate in the study. During inclusion, each amputee received a maximum of two written reminders. Non-responders who were admitted to our ward during the data collection period were asked by one of the authors (KØ) to participate in the study at the first day of their stay, unless they had refused to participate or had already received two written reminders.

The questionnaire was designed in consultation with The Norwegian Institute of Public Health and a physiotherapist and an occupational therapist from our ward with long experience in rehabilitation of ULAs. A figure of the arm was included in the questionnaire to clarify our definitions of amputation level.

Prior to data collection, our study was approved by the Norwegian Regional Ethics Committee. For the search for amputees in hospital medical records, we gained exemption from the duty of confidentiality from The Norwegian Directorate for Health and Social Affairs.

Variables

In the questionnaire study, data were collected on demographic features (gender, age at survey, ethnicity, marital status at survey, current place of

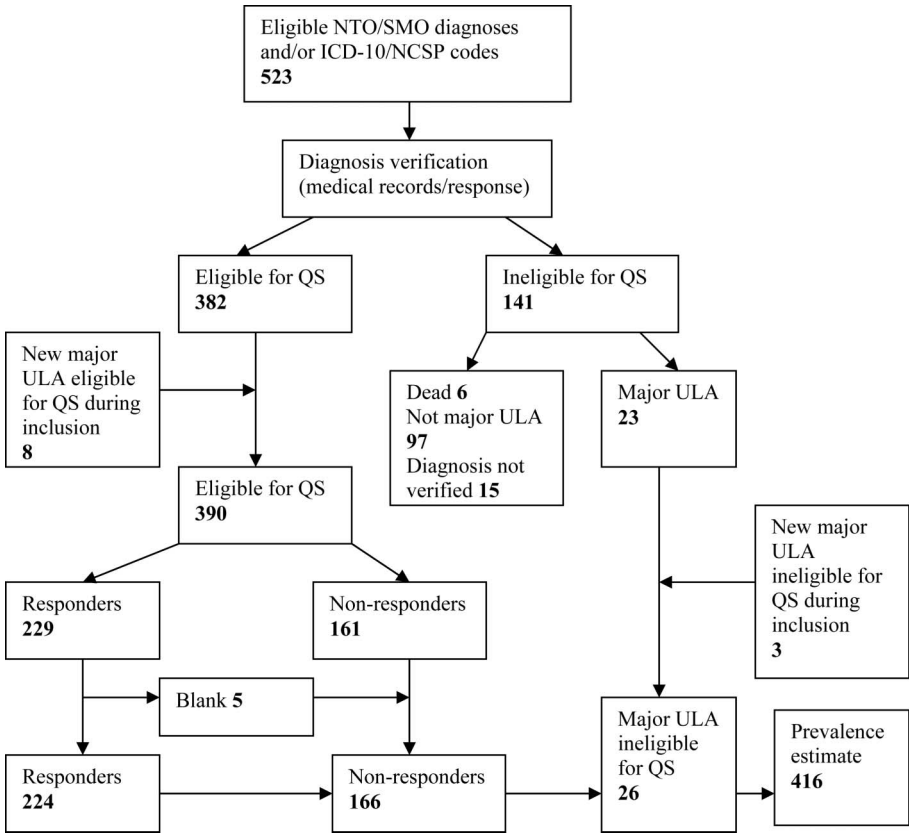


Figure 1. The inclusion process. Initially, 523 patients with eligible NTO/SMO-diagnoses and/or ICD-10 diagnostic and/or NCSP procedure codes were found. Control of medical records and responses to questionnaires left 141 patients ineligible for the QS. Of these, 23 were major ULA; 9 were excluded due to: insufficient Norwegian (5), severely reduced cognitive function or physical condition (2) and dementia (2), 12 questionnaires were returned due to unknown addressee (but according to DSP still resident in Norway) and 2 amputees had no permanent address. During inclusion, 3 additional major ULA ineligible for the QS were admitted to our ward; 2 did not master Norwegian (1 due to aphasia), 1 suffered from dementia. Initially, 382 major ULA were verified as eligible for the QS. During inclusion, another 8 new (recent) amputees were admitted to our ward and found eligible for the QS. In sum, we found 416 adult acquired major ULAs (prevalence estimate). Three hundred ninety were found eligible for the QS. Two hundred and twenty-four filled in questionnaires (responders). NTO, Norwegian Technical Orthopaedics AS; SMO, Sophies Minde Orthopaedics AS; QS, Questionnaire study; ULA, Upper limb amputees.

residence, occupational status at survey) and amputation specific features (age at amputation, side of amputation, level of amputation, geographic distribution, cause of amputation and current use of prostheses). Some background data were collected from the databases or medical records on all amputees (gender, age at survey, side and level of amputation). Current place of residence was found in the DSP. Ethnicity was assessed by name for all amputees and also by native country for the questionnaire responders.

Statistics

Data were processed using the Statistical Package for the Social Sciences (SPSS®) version 14.0, primarily by descriptive statistical analyses. 95% confidence intervals (CI) for percentages were calculated with a binomic formula. For hypothesis testing, the significance level was set at $\alpha = 0.05$. For categorical variables, a χ^2 test was performed to test for significant differences between groups. For continuous variables, a one-way ANOVA was performed.

Results

Estimation of population prevalence

We identified 416 major ULAs aged $\geq$18 years residing in Norway at the time of our study (see Figure 1). According to Statistics Norway, the Norwegian population per 1 January 2007 age 18–95 was 3,580,588 [47]. This yields an estimated population prevalence of 11.6 per 100,000. A 95% confidence interval is given by 10.5–12.7 per 100,000.

Population characteristics

In Table I, population characteristics are given for those variables known for all amputees in our

sampling frame, together with data on the same variables for the subgroups of responders, non-responders and amputees ineligible for the questionnaire study (QS). There were no significant differences regarding the variables examined.

Results from the questionnaire study

Two hundred twenty-four of 390 eligible amputees answered our questionnaires, yielding an overall 57.4% response rate. Demographic and amputation specific features of this selection are described below.

Demographic features

Data on gender, age at survey, ethnicity and place of residence are shown in Table I. We found an overall male:female ratio of 5:1. Nine of ten bilateral amputees were men. The majority of amputees in our sample were born in Norway. The rest were born in 17 different countries, mostly European. Of those born in Norway, only 6 [2.9% (95% CI = 0.6–4.8)] reported that one or both parents were not ethnically Norwegian. None of these parents were born outside Northern Europe. All Norwegian regions were represented. The pattern of residence was

Table I. Population characteristics.

	Group				
	Sampling frame	Questionnaire responders	Questionnaire non-responders	Ineligible for QS	$p^\star$ (two-sided)
n	416	224	166	26	NA
Gender %men (95% CI)	82.9 (79.3–86.5)	83.5 (78.6–88.3)	84.3 (78.8–89.9)	69.2 (51.5–87.0)	0.155
Age at survey (years)[†]					
Mean (95% CI)	53.8 (52.2–55.4)	53.7 (51.6–55.8)	53.7 (50.9–56.4)	55.2 (47.2–63.2)	0.912
Range	18.3–95.3	20.3–95.3	18.3–93.2	29.0–91.9	
Ethnicity based on name[‡]					
Number (% within group)					
Norwegian name	369 (88.7)	202 (90.2)	148 (89.2)	19 (73.1)	0.155
Nordic name	3 (0.7)	1 (0.4)	2 (1.2)	0	
Norwegian first name/non-Norwegian/ Nordic last name	6 (1.4)	2 (0.9)	3 (1.8)	1 (3.8)	
Non-Norwegian/Nordic name	38 (9.1)	19 (8.5)	13 (7.8)	6 (23.1)	
Ethnicity based on native country					
Number (%)					
Born in Norway	ND	202 (90.2)	ND	ND	
Born in other Nordic countries	ND	1 (0.4)	ND	ND	
Born outside the Nordic countries	ND	21 (9.4)	ND	ND	
Place of residence (region)[§]					
Number (% within group)					
East Norway	209 (50.2)	109 (48.7)	83 (50.0)	17 (65.4)	0.592
West Norway	105 (25.2)	54 (24.1)	44 (26.5)	7 (26.9)	
North Norway	40 (9.6)	24 (10.7)	16 (9.6)	0	
Trøndelag	36 (8.7)	20 (8.9)	15 (9.0)	1 (3.8)	
South Norway	26 (6.3)	17 (7.6)	8 (4.8)	1 (3.8)	
Unilateral/bilateral amputation					
Unilateral, *n*	392	214	152	26	0.286
% (95% CI)	94.2 (92.0–96.5)	95.5 (92.8–98.2)	91.6 (87.3–95.8)	100 (NA)	
Bilateral, *n*	21	10	11	0	
% (95% CI)	5.0 (2.9–7.2)	4.5 (1.8–7.2)	6.6 (2.8–10.4)	NA	
Unknown *n*	3	0	3	0	
% (95% CI)	0.7 (0.0–1.5)	NA	1.8 (0.2–3.8)	NA	
Side of amputation, unilateral amputees					
Right, *n*	211	112	88	11.	0.274
% (95% CI)	53.8 (48.9–58.8)	52.3 (45.6–59.0)	53.0 (45.4–60.6)	42.3 (23.3–61.3)	
Left, *n*	181	102	64	15	
% (95% CI)	46.2 (41.2–51.1)	47.7 (41.0–54.4)	38.6 (31.1–46.0)	57.7 (38.7–76.7)	
Level of amputation[¶]					
Forequarter					
n	13	8	4	1	0.092
% of known level within group (95% CI)	3.2[1] (1.5–4.9)	3.7[2] (1.2–6.2)	2.4[3] (0.1–4.8)	3.8 (0.0–11.2)	
Through shoulder joint					
n	23	7	13	3	
% of known level within group (95% CI)	5.6 (3.4–7.9)	3.2 (0.9–5.6)	7.9 (3.8–12.1)	11.5 (0.0–23.8)	

(*continued*)

Table I. (*Continued*).

	Group				
	Sampling frame	Questionnaire responders	Questionnaire non-responders	Ineligible for QS	$p^{\star}$ (two-sided)
Upper arm					
n	99	53	39	7	
% of known level within group (95% CI)	24.3 (20.1–28.4)	24.3 (18.6–30.0)	23.8 (17.3–30.3)	26.9 (9.9–44.0)	
Through elbow joint					
n	24	14	8	2	
% of known level within group (95% CI)	5.9 (3.6–8.2)	6.4 (3.2–9.7)	4.9 (1.6–8.2)	7.7 (0.0–17.9)	
Forearm					
n	176	85	81	10	
% of known level within group (95% CI)	43.1 (38.3–47.9)	39.0 (32.5–45.5)	49.4 (41.7–57.0)	38.5 (19.8–57.2)	
Through wrist					
n	73	51	19	3	
% of known level within group (95% CI)	17.9 (14.2–21.6)	23.4 (17.8–29.0)	11.6 (6.7–16.5)	11.5 (0.0–23.8)	
Unilateral, unknown level (n)	3	1	2	0	
Bilateral, not same level on both sides (*n*)**	5	5	0	0	

QS, Questionnaire study; CI, Confidence interval; ND, No data; NA, Not applicable.

*Comparison between groups by either Chi square test (for frequencies) or one-way ANOVA (for continuous variables).

†Age at survey is given as age at response for responders, and as age at the time of mailing questionnaires for non-responders and those ineligible for QS. For those ineligible not considered for questionnaires, age is given as age at the time of admission to our ward ($n = 4$) or as age in January 2007 (when most questionnaires were sent out) ($n = 2$).

‡Ethnicity is given based on name because native country is known only for responders. As shown, it is good correspondence between name and native country in the responder group.

§Norway is divided into five geographic regions based on county; East Norway: Østfold, Akershus, Oslo, Hedmark, Oppland, Buskerud, Vestfold, Telemark. West Norway: Rogaland, Hordaland, Sogn og Fjordane, Møre og Romsdal. North Norway: Nordland, Troms, Finnmark. Trøndelag: Sør-Trøndelag, Nord-Trøndelag. South Norway: Aust-Agder, Vest-Agder. For amputees with unknown address at the time of the survey ($n = 14$), the last known place of residence is given.

¶Bilateral amputees with the same level on both sides included (5 responders, 11 non-responders). Bilateral responders: 2 through wrist, 2 fore arm, 1 upper arm.

**Bilateral responders: 2 through wrist + forearm, 1 through wrist + through elbow joint, 1 forearm + through shoulder joint, 1 through elbow joint + through shoulder joint.

[1]% of 408; [2]% of 218; [3]% of 164.

concurrent with the number of residents in each region [48].

Two hundred twenty-two amputees reported their marital status at the time of the survey. Fifty-nine percent (95% CI = 52.5–65.5) were married, 11.7% (95% CI = 7.5–15.9) cohabitant, 7.2% (95% CI = 3.8–10.6) divorced, 18.9% (95% CI = 13.8–24.1) single and 3.2% (95% CI = 0.9–5.5) widowed. Occupational status at survey is shown in Table II.

Table II. Occupational status at survey among questionnaire responders ($n = 224$).

Occupational status	*n*	%	95% CI
Work at home	4	1.8	0.05–3.5
Student	8	3.6	1.1–6.0
Employed	68	30.4	24.3–36.4
Self-employed	17	7.6	4.1–11.1
Disability pension	72	32.1	26.0–38.3
Retired	45	20.1	14.8–25.3
Unemployed	3	1.3	0.0–2.8
Occupational rehabilitation	7	3.1	0.8–5.4
Total	224	100.0	NA

NA, Not applicable.

Amputation specific features

Data on unilateral/bilateral distribution, side of amputation and level of amputation are shown in Table I. Most ULAs were unilateral. There was no predominant side of unilateral amputation and there was no significant tendency towards more frequent amputation of the dominant hand (χ^2 test: $p = 0.61$). Among unilateral amputees, 31.5% (95% CI = 25.2–37.7) were amputated proximal to the elbow joint, 45.6% (95% CI = 38.9–52.2) were amputated through the elbow joint or forearm and 23.0% (95% CI = 17.4–28.7) were amputated through the wrist.

Age at amputation ($n = 223$). Mean age at amputation was 29.6 years (95% CI = 27.4–31.9), range 0–80 years. Seventy-five percent of the amputees were ≤36.1 years at amputation, 25% ≤14.3 years.

There were no significant gender differences in age at amputation. Mean time passed since the amputation was 24.0 years (95% CI = 21.8–26.3).

Cause of amputation ($n = 220$). The dominant causes of upper limb loss in our sample were

traumatic, trauma accounting for 84.5% of unilateral amputations and nine of ten bilateral amputations. Occupational accidents and traffic accidents were most frequent. Details are given in Table III. There were statistically significant gender differences in cause of amputation in our sample (χ^2 test: $p < 0.001$). Women had a higher prevalence of amputation due to traffic accidents than men, whereas men had a higher prevalence of amputation due to occupational accidents than women. Amputation due to war injury, explosion, torture, self-inflicted damage and overdose occurred solely among the men in our sample.

The main causes of amputation varied according to amputation level (χ^2 test: $p < 0.001$). Details are given in Table IV. Trauma was the major cause of amputation distal to the shoulder joint. At the shoulder and forequarter level cancer was the leading cause of amputation. 53.3% of upper limb amputations due to cancer were performed at the shoulder or forequarter level. All upper limb amputations due to explosions were performed through the wrist or through the forearm.

Amputation country and hospital. Two hundred five [91.9% (95% CI = 88.4–95.5)] of the 224 amputees in our sample were amputated in Norway. Another 18 were amputated in 14 different countries all over the world. One hundred ninety-six amputees amputated in Norway gave information about the hospital where the amputation took place. They were amputated at 52 different hospitals, geographically distributed all over Norway. The hospitals with the highest number of amputations were the largest of the Norwegian hospitals, including Haukeland University Hospital with 18 amputations and St. Olavs Hospital with 14 amputations. In Oslo, the hospitals have been reorganised several times over the years. In sum, 59 amputees were amputated at seven different hospitals in Oslo from 1942 until 2007.

Traumatic amputation - time delay - reimplantation. Among 177 traumatic amputees and two amputees due to overdose, 102 (57.0%) were amputated the same day as the trauma/overdose. Nine more were amputated the following day. Within 1 month 146 (81.6 %) were amputated and within a year the number of amputations was 170 (95.0 %). The last nine were amputated 1.5–24.5 years after the trauma, seven of these within 4.5 years. One hundred eighty traumatic amputees gave information on whether reimplantation had been attempted. This was the case for only 23 amputees [12.8% (95% CI = 7.9–17.7)].

Current use of prostheses. Two hundred nine of our 224 responders [93.3% (CI = 90.0–96.6)] stated that they had worn prostheses at some time after the amputation. Eleven had never worn prostheses [4.9% (95% CI = 2.1–7.7)]. Two hundred eight amputees gave information on current use of prostheses. Twenty-nine of these [13.9% (95% CI = 9.2–18.6)] were no longer wearing prostheses. The current prostheses-wearing proportion of the amputees was 179/224 [79.9% (95% CI = 74.7–85.2)].

Table III. Cause of amputation among questionnaire responders ($n = 220$).

Cause	Total (n)	% of total	95% CI of total%	Men (n)	% of men	95% CI of % men	Women (n)	% of women	95% CI of % women
Occupational accident	91	41.4	34.9–47.9	87★	46.8	39.6–53.9	4	11.8	0.9–22.6
Traffic accident	31	14.1	9.5–18.7	17	9.1	5.0–13.3	14	41.2	24.6–57.7
War injury	10	4.5	1.8–7.3	10†	5.4	2.1–8.6	0	0	NA
Explosion	19	8.6	4.9–12.3	19‡	10.2	5.9–14.6	0	0	NA
High voltage	5	2.3	0.3–4.2	3§	1.6	0.0–3.4	2	5.9	0.0–13.8
Other accidents	28	12.7	8.3–17.1	22	11.8	7.2–16.5	6	17.6	4.8–30.5
Torture	1	0.5	0.0–1.3	1	0.5	0.0–1.5	0	0	NA
Self-inflicted damage	1	0.5	0.0–1.3	1	0.5	0.0–1.5	0	0	NA
Cancer	15	6.8	3.5–10.1	11	5.9	2.5–9.3	4	11.8	0.9–22.6
Infection	14	6.4	3.1–9.6	11	5.9	2.5–9.3	3¶	8.8	0.0–18.3
Arteriosclerosis/poor circulation/diabetes	3	1.4	0.0–2.3	2	1.1	0.0–2.6	1	2.9	0.0–8.6
Overdose	2	0.9	0.0–2.2	2	1.1	0.0–2.6	0	0	NA
Total	220	100.0	NA	186	100.0	NA	34	100.0	NA

NA, Not applicable.
★2 bilateral included.
†3 bilateral included.
‡2 bilateral included.
§2 bilateral included.
¶1 bilateral included.

Table IV. Cause in relation to level of amputation among unilateral responders ($n = 210$).

	Level of amputation: *n* (% within cause)						
Cause of amputation	Forequarter	Through shoulder joint	Upper arm	Through elbow joint	Forearm	Through wrist	Total
Occupational accident	1 (1.1)	3 (3.4)	18 (20.2)	6 (6.7)	41 (46.1)	20 (22.5)	89 (100.0)
Traffic accident	1 (3.2)	0	19 (61.3)	2 (6.5)	8 (25.8)	1 (3.2)	31 (100.0)
War injury	0	0	0	1 (14.3)	2 (28.6)	4 (57.1)	7 (100.0)
Explosion	0	0	0	0	6 (35.3)	11 (64.7)	17 (100.0)
High voltage accident	0	0	0	1 (33.3)	2 (66.7)	0	3 (100.0)
Other accident	0	1 (3.6)	5 (17.9)	1 (3.6)	11 (39.3)	10 (35.7)	28 (100.0)
Self-inflicted damage	0	0	0	0	0	1 (100.0)	1 (100.0)
Torture	0	0	1 (100.0)	0	0	0	1 (100.0)
Infection	1 (7.7)	0	5 (38.5)	2 (15.4)	5 (38.5)	0	13 (100.0)
Cancer	5 (33.3)	3 (20.0)	3 (20.0)	0	4 (26.7)	0	15 (100.0)
Arteriosclerosis/poor circulation/diabetes	0	0	0	1 (33.3)	2 (66.7)	0	3 (100.0)
Overdose	0	0	1 (50.0)	0	1 (50)	0	2 (100.0)
Total	8 (3.8)	7 (3.3)	52 (24.8)	14 (6.7)	82 (39.0)	47 (22.4)	210 (100.0)

Discussion

Validity of our prevalence estimate

Prevalence estimates are sensitive to selection errors. Norway has a well organised national public health service, with high quality systems for registration of patients and diagnoses. This makes an estimation of prevalence feasible – although miscoding does occur and must be remembered as a source of error. In our study, we performed a thorough control of all major ULA diagnoses, thus confirming their validity. The databases were explored both electronically and manually, minimising the risk of missing data. Also, the lack of additional cases after cross-checking against amputation-related diagnostic codes and procedure codes suggests that our search may be comprehensive. Most amputees were registered more than once at the same hospital, and many amputees were registered at more than one hospital and/or in the databases of both prosthetic suppliers. The possibility to pool the lists and remove duplicates based on individual National Insurance numbers gave us a unique possibility to obtain a comprehensive list of Norwegian adult acquired major ULAs.

We may however have missed some amputees due to the 'time limit' of the electronic databases of the Norwegian prosthetic suppliers; the electronic records starting in 1994 (NTO AS) and 1999 (SMO AS), respectively. Also, the Norwegian version of the ICD-10 was implemented for somatic diseases from 1 January 1999. Major ULAs amputated before this and not in contact with a prosthetic supplier during these years will have been missed. Due to our inclusion procedures, major ULAs not registered in the DSP will also have been missed. Furthermore, upper limb amputations are performed in all Norwegian hospitals and some non-users will have been missed due to our search for non-users in only three large hospitals. However, the literature suggest that non-users constitute only 1.5–6% of the total upper limb amputee population [6,49–51], making this a minor source of error. In sum, our study is not 100% comprehensive and our prevalence estimate is likely to be conservative.

Limitations of the questionnaire study

A relatively low response rate is a weakness of the questionnaire study. This may have lead to selection bias in our sample. But, as shown in Table I, the responders did not differ substantially from the non-responders or from the sampling frame regarding the demographic and amputation specific features examined. This suggests that the bias may not have seriously affected the results.

Though inclusion of patients from large hospitals in three of five Norwegian geographic regions were done in order to find a representative sample of non-users, our inclusion procedures may have caused a selection bias that may have affected the results of the questionnaire study. However, the geographic distribution of the sample frame and the responders throughout the entire country suggests that overall, no regions were underrepresented.

Finally, interpreting the results of the questionnaire study one should take into consideration the possibility of recall bias generally associated with the cross-sectional design. Also, applying the QS-specific inclusion and exclusion criteria was in some cases difficult due to missing information. In such cases, the amputee may have been wrongfully included in

the QS, possibly introducing missing data due to lack of ability to give the information required.

Comparison of our prevalence estimate to previous studies

Earlier publications estimating the prevalence or the incidence of upper limb loss in the Nordic and other western countries are summarised in Table V. As shown, population-based studies are scarce and differences in methodology and study population make comparisons difficult. Our prevalence estimate does however correspond to the number of users of arm prostheses found by SINTEF Unimed Rehab in 1996 [21] and to the prevalence estimate of acquired major upper limb loss in the USA found by Ziegler-Graham in 2005 [22]. The inclusion of children and partial hand amputees in the SINTEF report implies that the number of adult acquired major ULAs in Norway at the time of our survey might be slightly increased compared to 1996. This assumption is also supported by the fact that the SINTEF report did not correct for duplicates. In our survey, such duplicates registered at both prosthetic suppliers constituted about 3.5%. On the other hand, the SINTEF report was not comprehensive, missing an estimate of the number of non-users. In comparison, 13.3% of the sampling frame in our survey was potential non-users not registered in the databases of NTO AS or SMO AS.

Incidence rates

As shown in Table V, earlier incidence estimates vary considerably. It was beyond the scope of this survey to estimate the incidence of adult acquired major upper limb loss in Norway. A rough estimate may however be deduced from the 11 new adult major ULAs who were registered during our 20 months of inclusion (see Figure 1), yielding an incidence estimate of 0.12 per 100,000 per year, or 6.6 new major amputees per year. However, this estimate only represents new amputees who were referred to our ward during inclusion. Other new amputees will have been treated and/or supplied with prostheses elsewhere. It is nevertheless interesting that this incidence estimate is very similar to the one presented by Atroshi and Rosberg in 2001; 0.11 per 100,000 person years [20], and also in accordance with the estimate of 0.08 per 100,000 per year ($n = 7$) made by Nylander et al. in 1984/85 [17,18]. Our incidence estimate is considerably lower than the one given by SINTEF Unimed Rehab in 1995; 30 acquired arm amputees per year [21]. The basis of the SINTEF estimate is however somewhat unclear. Nevertheless, a genuine decline in incidence over time would be in accordance with the literature. In USA and in England such a downwards trend has been noted over the last 20 years, explained by better safety in the workplace, a decline in heavy industry (exposing fewer individuals to risk of injury) and increasing replantation success [16,21,36].

Demographic features

The amputees in our sample were predominantly men amputated at relatively young age, consistent with earlier studies on acquired ULAs [2,3,5–7, 10,13,14,21,27,52,54–57]. Their marital status at survey was also similar to findings in other studies [3,7,39]. Compared to the Norwegian general population at the time of the survey, the percentage of married amputees was significantly higher and the percentage of single amputees was significantly lower. Some of the difference in marital status may be explained by age; the mean age of our study population was higher than in the general population possibly explaining a higher marital rate than among younger Norwegians. However, a higher marital rate and a lower divorce rate among ULAs than in the general population has also been noted in Denmark by Kejlaa, who explained the low divorce rate by a symbiosis between the amputees and their partners to protect their future existence [3].

The percentage of ULAs in paid employment in Norway was relatively low, but of about the same magnitude as found elsewhere [10,13,57]. Referring the literature, Davidson in 2002 noted that between 70% and 90% of amputees return to work at some time after their injuries, but the proportion of amputees who maintain employment for a long period drops to between 40 and 75% [13]. This is in accordance with our findings, mean post-amputation time in our sample being 24 years. The relatively high percentage of retired amputees in our sample can be explained by the mean age in our sample approaching normal retirement age: in Norway one can retire due to age from 62.

Most amputees in our sample were ethnically Norwegian, amputated in Norway. Refugees amputated due to war trauma constituted a smaller proportion than we expected from clinical experience. A plausible explanation is that we see a selective subgroup of amputees in need of extensive multidisciplinary rehabilitation as inpatients in our ward.

Amputation specific features

The number of bilateral amputees in our material was of the same magnitude as reported elsewhere [8,11,28,53,58–60]. In accordance with the majority of the literature [2,5–7,10,12,16,21,35,54,56,61–63]

Table V. Earlier publications estimating the incidence or the prevalence of upper limb loss in the Nordic and other western countries.

Author	Publication year	Country	Prevalence estimate	Incidence estimate	Study population/ Data source
Nordic countries					
SINTEF Unimed Rehab report [21]	1996	Norway	417 registered users of arm prostheses	30 acquired arm amputations per year	Upper limb prosthetic users registered at NTO AS or SMO AS. Acquired amputees. Partial hand amputees and children included. Duplicates not removed. The basis of the incidence estimate is somewhat unclear.
Atroshi and Rosberg [20]	2001	Norway and Sweden	–	0.11 per 100,000 person-years	2 epidemiologic studies concerning hand injuries, one of which was conducted in three Norwegian cities, constituting approximately 5% of the Norwegian population. Incidence estimate of major ULA in Norway, all ages (children included) for the 9 year period from 1991 to 1999.
Nylander, Villki and Östrup [17,18]	1984/85	Sweden	–	7 per year (0.08 per 100,000 per year)	All work-related registered serious amputation injuries of the upper extremity in Sweden during 1979 were investigated. The number of major amputations was 7 in a population of 8 million. Ages 10–69 included.
Andersen-Ranberg and Ebskov [19]	1988	Denmark	–	75 per year 150 per 100,000 per year	Upper extremity amputations (not amputees) in Denmark 1978–83, based on the Danish Amputation Register. All ages. Partial hand amputations included.
Pohjolainen and Alaranta [52]	1988	Finland	–	20–30 cases per year 0.3 amputations per 100,000 inhabitants per year	Estimate of the annual number of major upper limb amputation in Finland in 1984–85, congenital amputations included. Estimates based on a study of the catchment area of the Helsinki University Central Hospital corresponding to 24% of the population of Finland.
Other western countries					
Ziegler-Graham et al. [22]	2005	USA	41,000 (13.5 per 100,000)	–	Estimate of the number of acquired major upper limb amputees in the USA, ages 0–85 years included. The prevalence estimate per 100,000 has been deduced from US population information given in the paper.
Dillingham et al. [36]	2002	USA	–	0.58 per 100,000 US population (1988) 0.54 per 100,000 US population (1996)	Examined incidence rates of limb amputations in the USA using nationally representative hospital discharge data from 1988 to 1996. Incidence rates for acquired major upper limb amputation (through or proximal to the wrist) due to trauma, cancer or dysvascular cause are extracted. All ages included. 1996 rates are standardised to the 1988 US population by age, sex and geographic region.
Jones and Davidson [53]	1997	New South Wales, Australia	–	0.56 per 100,000	Review. Statistics from New South Wales Department of Health 1993. Acquired major upper limb amputations. Partial hand included? Ages not specified.
Van der Sluis et al [14]	2009	The Netherlands	–	0.26 per 100,000 inhabitants	Refers to Dutch statistics for estimation of incidence of upper limb amputations, age and level not specified. (The study looked at major amputations ages 18–65 registered at a large European University Medical Centre during the last two decades.)

(*continued*)

Table V. (*Continued*).

Author	Publication year	Country	Prevalence estimate	Incidence estimate	Study population/ Data source
Chappel [27]	1992	England	–	239 single arm amputations per year 3 double arm amputations per year	Refers to arm amputation statistics for England 1958–88 by the Great Britain Department of Health. The median annual number of patients with acquired arm amputations is given, all ages included, level not specified (fingers and/or partial hand may be included).
UK prosthetic services [28]	2007	UK	–	135 per year	National Amputee Statistical Database for the UK. Total number of new major acquired arm amputee referrals to UK prosthetic services centres in 2005/ 2006, all ages included.

NTO AS, Norwegian Technical Orthopaedics AS; SMO AS, Sophies Minde Orthopaedics AS. These are the two only companies in Norway that make arm prostheses.

there was no predominant side of unilateral amputation and no significant tendency towards more frequent amputation of the dominant extremity. The percentage of prosthetic users was in accordance with recent literature, though documented rates of prosthetic use among ULAs vary considerably, between 27 and 81% [3,5,11,39,53,55,58,64].

The distribution of amputation level was also in agreement with the majority of other statistics, the number of forearm and wrist amputees exceeding that of more proximal amputees [2,7–13,23,35,38, 55,56,60,63,65–68]. This distribution is most prominent when looking at traumatic amputees only [9,11]. In our material, 23.4% of the responders were amputated through the wrist. With a few exceptions [2,9], this is a much higher percentage of wrist disarticulation than has been reported elsewhere. In studies where a segregation between transradial amputation and wrist disarticulation has been made, the percentage of wrist disarticulation has been reported between 1.5% and 14.3 %, most often around 7–8% [6–8,12,13,28,35,39,53,56,64]. This corresponds better to the percentage of wrist disarticulation among our non-responders. One possible explanation is that we have a selection bias in our material, with a higher prevalence of wrist disarticulation among the responders. But also in our selection frame, the proportion of through-wrist amputees is larger than seen elsewhere. In Norway, NTO AS specialises in making prostheses and orthoses for the improvement of grip, also for wrist or distal forearm amputees. This may have led to a higher registered proportion of distal major amputees in our material compared to other countries, where the use of prostheses by very distal major amputees may be less common.

In accordance with the literature, the dominant causes of acquired limb loss in our sample were traumatic [2,3,5–7,11–14,16,22,27,28,35,39,53,55, 58,60,62,68,69]. Occupational accidents were most frequent, followed by traffic accidents. This trend is also seen elsewhere [3,10,27,40,52]. The distribution of causes in relation to level of amputation was as expected from our clinical experience and also in accordance with the literature, accidents being the most frequent cause of distal amputation and malignancy the dominant cause of the most proximal amputations [19,28,70]. Interestingly, there were statistically significant gender differences in cause of amputation in our sample; women had a higher percentage of amputation due to traffic accidents, whereas men had a higher percentage of limb loss due to occupational accidents. Earlier, a high male:female ratio in trauma-related amputation has been noted [36]. Otherwise, literature on gender differences in amputation cause is scarce. Our findings might be explained by the hypothesis that for women, the traffic is the most likely place to get injured, assuming that women choose occupations and leisure activities with less risk of trauma than men do.

Some post-traumatic amputations were very delayed in time after the injury. These amputations may be due to sequelae in the form of nerve damage, leaving the injured arm painful and/or without function.

Conclusions

Our estimate of prevalence of adult acquired major upper limb amputation in Norway is 11.6 per 100,000 adults ($n = 416$). It is likely that this estimate is conservative. The amputees were predominantly men with traumatic, unilateral, distal amputations at a young age. About four in ten were in paid

employment. More than nine in ten had used prostheses after the amputation.

Literature on ULAs – especially population-based studies and prevalence estimates – is scarce, our study therefore being an important addition to existing knowledge. Our description of the demographic and amputation specific features of the Norwegian major ULA population is mainly consistent with earlier studies from other countries, implying that existing knowledge – with caution regarding methodological issues – may be applied in the planning of future healthcare in our country and that our findings may be applicable in other western countries. Earlier incidence estimates vary considerably, and the authors suggest that a comprehensive prospective population-based study is carried out in order to estimate the incidence of major ULA in Norway. Furthermore, the establishment of a registry of Norwegian acquired ULAs should be considered. Our data may form the basis of such an amputation registry, which might become a valuable tool in monitoring, planning and carrying out future amputee rehabilitation.

According to our work, the number of ULAs is relatively small, indicating that rehabilitation should be carried out at a few sites, allowing necessary expertise to be built up and maintained. This is in accordance with the current regional organisation of ULA rehabilitation in Norway. Whether the amputees should be offered inpatient and/or outpatient rehabilitation and which professions should be represented in the multidisciplinary teams will be explored in a further study of ULA rehabilitation and functional outcome. At present, this varies from site to site. The relatively low percentage of ULAs in paid employment indicates that occupational rehabilitation should be emphasised. Furthermore, the geographic distribution of upper limb amputations indicates the importance of all Norwegian hospitals being informed about the rehabilitation sites available. To ensure optimal rehabilitation, referral to a rehabilitation unit should probably be mandatory at discharge from the surgical ward.

The number of traumatic amputations indicates that preventive measures may reduce future amputation incidence. The frequency of occupational accidents indicates that there is still work to be done regarding safety in the workplace. Useful information on where to allocate resources might arise from a detailed study of amputation causes for instance exploring which workplaces are the most prone to accidents.

A high proportion of through-wrist amputees in Norway may be related to patterns of prosthetic use and registration and does not necessarily reflect a true difference from other countries. Nevertheless, the number of distal amputees needs to be taken into consideration by the Norwegian health care system, especially with regards to the capacity of the prosthetic services. The pattern of prosthetic use among these and other Norwegian major ULAs will be investigated in a further study.

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